# Hyun Yoon

Postdoctoral scholar, **University of California, Berkeley**

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**Research Profile:** I develop predictive frameworks for contaminant fate and transformation by integrating environmental geochemistry, environmental microbiology, and data-driven modeling.

## EDUCATION

Ph.D. **Cornell University**, Ithaca, New York — 2019–2023
Civil and Environmental Engineering

M.S. **Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, South Korea — 2016–2018
Civil and Environmental Engineering

B.S. **Yonsei University**, Seoul, South Korea — 2010–2016
Atmospheric Sciences
*- Exchange Program,* ***University of California, Los Angeles*** *in 2014 (1 year)*

## RESEARCH EXPERIENCE

Postdoctoral Scholar, **University of California, Berkeley** — 2024–present
& NIEHS Superfund Research Program Trainee
- Mentors: Profs. Lisa Alvarez-Cohen & David L. Sedlak
- *Research focus*: Environmental microbiology and geochemistry governing contaminant fate and transformation, with a primary focus on PFAS

Graduate Research Assistant, **Cornell University** — 2019–2023
- Advisor: Prof. Matthew C. Reid
- *Research focus*: Wetland biogeochemistry with a focus on arsenic and carbon dynamics

Graduate Research Assistant, **KAIST** — 2016–2018
- Advisor: Prof. Sukhwan Yoon
- *Research focus*: Biological treatment of $N_2O$ emissions from wastewater treatment plants

## HONORS AND AWARDS

National Interest Waiver (NIW), **United States Citizenship and Immigration Services** — 2025
(U.S. permanent residency granted based on publication record and expert references)

International Research Travel Grant, **Cornell University** — 2022
(Proposal selected, funding two-month visit to uOttawa for on-site collaboration)

Featured in Journal Spotlight, ***Applied and Environmental Microbiology*** — 2022
(Selected by the editorial team in the respective issue)

Cornell Fellowship, **Cornell University** — 2019–2020
(Competitively awarded full funding package for the first year of Ph.D. study)

KAIST Support Scholarship for State Scholarship Student, **KAIST** — 2016–2018
(Competitively awarded full funding package for the duration of the M.S. program)

Internal Scholarship & Academic Awards, **Yonsei University** — 2015
(Competitively awarded final semester tuition funding)

Excellence Award, **Republic of Korea Army** Squad Leader Training Course — 2012
(Awarded for ranking 2nd overall among ~200 trainees)

## PUBLICATION RECORD

*Peer-reviewed Journal Publications:*

1. Sarah Choyke[†], **Hyun Yoon**[†], Caroline Rose Alukkal, Youn Jeong Choi, Collin Meyers, Jonathan Thorn, Hannah Kruelle, Anthony Grimaldi, Ben Priest, Taryn McKnight, Josie King, Ananya Visweswaran, James Gray, Linda S. Lee, Lisa Alvarez-Cohen, David L. Sedlak, Fuhar Dixit and Mohamed Ateia. 2026. Versatile TOP Assay Surrogates for Evaluating Oxidation Efficiency in Complex Environmental Samples. ***Journal of Hazardous Materials Letters***. 7. 100198. ***([†]Equal first author contribution)***
2. Fuhar Dixit, Sarah Suhnholz, **Hyun Yoon**, Elizabeth M. Fries, Edmund H. Antell, Christopher P. Higgins, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Evaporative losses of perfluoroalkyl sulfonamides during PFAS analysis. ***ACS Omega,*** 11(9), 14638-14643.
3. Elisa Garabello, **Hyun Yoon**, Matthew C. Reid and Andrea Giometto. 2026. Tunable low-rate genomic recombination with *cre-lox* in *e. coli*: a versatile tool for synthetic biology and environmental sensing. ***Applied and Environmental Microbiology***, 92(4), e1768-25.
4. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Effects of organic matter, iron (hydr)oxides, and iron reductive dissolution on PFAS sorption in AFFF-impacted solids. ***Environmental Science & Technology Letters,*** 13(1), 130-136.
5. **Hyun Yoon**, Michael Vega, Jiaxing Wang, Alexandre J. Poulain, Andrea Giometto, Ludmila Arstilde and Matthew C. Reid. 2024. Repression of arsenite uptake and methylation by dissolved organic matter. ***Environmental Science & Technology Letters***, 11(8), 838-844.
6. Heejoo Han, Daehyun D. Kim, Min Joon Song, Taeho Yun, **Hyun Yoon**, Hong Woon Lee, Young Mo Kim, Michele Laureni and Sukhwan Yoon. 2023. Biotrickling filtration for the reduction of $N_2O$ emitted during wastewater treatment: results from a long-term *in situ* pilot-scale testing. ***Environmental Science & Technology,*** 57(9), 3883-3892.
7. **Hyun Yoon**, Benjamin Stenzler, Lena Abu-Ali, Maria P. Asta, Alexandre J. Poulain and Matthew C. Reid. 2023. Effects of iron and dissolved organic matter on bioavailability of arsenite under anaerobic conditions. ***ACS ES&T Water***, 3(11), 3676-3686.
8. **Hyun Yoon**, Andrea Giometto, Martin P. Pothier, Xuhui Zhang, Alexandre J. Poulain and Matthew C. Reid. 2022. Time-dependent biosensor fluorescence as a measure of bacterial arsenic uptake kinetics and its inhibition by dissolved organic matter. ***Applied and Environmental Microbiology***, 88(16), e00891-22. **Featured in *AEM* Research Spotlight**
9. Lena Abu-Ali, **Hyun Yoon** and Matthew C. Reid. 2022. Effects of organic sulfur and arsenite/dissolved organic matter ratios on arsenite complexation with dissolved organic matter. ***Chemosphere***, 302, 134770.
10. Daehyun D. Kim, Doyoung Park, **Hyun Yoon**, Taeho Yun, Min Joon Song and Sukhwan Yoon. 2020. Quantification of *nosZ* genes and transcripts in activated sludge microbiomes with novel group-specific qPCR methods validated with metagenomic analyses. ***Water Research***. 185, 116261.
11. Min Joon Song, **Hyun Yoon**, Tae Kwon Lee, Miye Kwon and Sukhwan Yoon. 2020. Orthophosphate enhances abiotic nitrous oxide production from aerobic hydroxylamine decomposition. ***Biology and Fertility of Soils,*** 56(5), 687-695.
12. **Hyun Yoon**, Min Joon Song, Daehyun D. Kim, Fabrizio Sabba and Sukhwan Yoon. 2019. A serial biofiltration system for effective removal of low-concentration nitrous oxide in oxic gas streams: mathematical modeling of reactor performance and experimental validation. ***Environmental Science & Technology,*** 53(4), 2063-2074.
13. **Hyun Yoon**, Min Joon Song and Sukhwan Yoon. 2017. Design and feasibility analysis of a self-sustaining biofiltration system for removal of low concentration $N_2O$ emitted from wastewater treatment plants. ***Environmental Science & Technology***, 51(18), 10736-10745.

*Selected Manuscripts under review and in preparation:*

1. **Hyun Yoon**[*], John Han, Lisa Alvarez-Cohen and David L. Sedlak. Molecular structure and environmental context jointly predict PFAS biotransformation outcomes: a literature-derived machine learning analysis. *In prep.* ***([*]Correspondence)*** [Research Poster Link]

2. Anna-Ricarda Schittich[†], **Hyun Yoon**[†], Leyan Peng, Fuhar Dixit, Tae-Kyoung Kim, Olivia Wang, Fanshu Geng, Damian Helbling, Christopher Olivares, David L. Sedlak and Lisa Alvarez-Cohen. Charge-dependent sorption controls the bioavailability and biotransformation of PFAS precursors in AFFF-impacted solids. *In prep.* ***([†]Equal first author contribution)*** [Research Poster Link]
3. **Hyun Yoon**[†], Michael Vega[†] and Matthew C. Reid. Genome-resolved metatranscriptomic analysis of arsenic demethylation and detoxification in a methanogenic rice paddy soil. ***bioRxiv***, 2026-08 [Link] ***([†]Equal first author contribution)***
4. Sarah Suhnholz, Boris Droz, **Hyun Yoon**, Corey De La Cruz, Nayantara T. Joseph, Johanna Peeters Weem, Trevor Schwichtenberg, Eric Redman, Mark Benotti, Gerrad D. Jones, Damian E. Helbling, Lisa Alvarez-Cohen, David L. Sedlak, Jennifer A. Field, Christopher P. Higgins and Fuhar Dixit. PFAS forensics strategy for source identification: an intermethod comparison across environmental challenge matrices. *In prep.*

*Technical Reports & Government Documents:*

1. Fuhar Dixit, **Hyun Yoon**, Lisa Alvarez-Cohen and David L. Sedlak, 2026. A simple and robust forensic technique for differentiating PFAS associated with AFFF from other PFAS sources. ***SERDP Final Report***, ER20-1330. [Link]

## GRANT ACTIVITIES

*Project 3 Proposal Lead*, **UC Berkeley Superfund Research Program (NIEHS P42)** (USD 3.8M proposed) — 2026 (*Under Review*)

- Title: "PFAS Release from Long-Term Sources and Their Fate in Subsurface Environments" (FY2027–2032 Renewal)
- Role: Spearheaded the proposal development of engineering portion (Project 3) for the 5-year center renewal. Authored technical progress reports of the previous term, synthesized preliminary fate/transformation data, and actively collaborated with biomedical teams (Projects 1 & 2) to integrate engineering outcomes with biological exposure pathways across campus.

*Proposal Lead*, **U.S. Department of Defense, SERDP** (USD 1.7M proposed) — 2025 (*Submitted/ not funded*)

- Title: "Effect of Subsurface Solids Properties and Redox Conditions on PFAS Transformation at AFFF-Impacted Sites"
- Role: Led the development of the pre-proposal and full proposal.

*Postdoctoral Project Lead,* **U.S. Department of Defense, SERDP** (USD 1.8M) — 2024–2027 (*Funded*)

- Title: "Assessment of Physical, Chemical, and Biological Factors Controlling Biotransformation of Cationic and Zwitterionic Precursors in PFAS Source Zones (ER23-3827)"
- Role: Led day-to-day research implementation and project coordination, including experimental planning, data collection and analysis, budget management, mentoring of team members, progress presentations and reporting, and coordination with project collaborators.

*Postdoctoral Project Lead*, **U.S. Department of Defense, SERDP** (USD 1.5M) — 2021–2025 (*Funded*)

- Title: "A Simple and Robust Forensic Technique for Differentiating PFAS Associated with AFFF from Other PFAS Sources (ER20-1330)"
- Role: Led day-to-day research activities and project coordination, including data collection and analysis, mentoring of team members, progress reporting, and collaboration across participating research groups.

*Principal Investigator*, **International Research Travel Grant, Cornell University** (USD 3K) — 2022 (*Funded*)

- Title: "Effects of Iron and Organic Matter on Arsenic Bioavailability Using Novel Anaerobic Biosensor Techniques"

- Role: Developed the full proposal and conducted a two-month research collaboration with Prof. Alexandre Poulain's group at the University of Ottawa.

*Graduate Researcher / Project Contributor*, **NSF Environmental Chemical Sciences Program, Division of Chemistry** (USD 355K) 2019–2022 (*Funded*)
- Title: "Arsenic Complexation with Reduced Organosulfur Moieties in Soil Organic Matter: Implications for Arsenic Oxidation via Biotic and Abiotic Pathways"
- Role: Post-award team member, data collection and analysis, collaborative partnerships.

*Graduate Researcher / Project Contributor*, **USDA-NIFA Bioenergy, Natural Resources, and Environment Program** (USD 495K) 2018–2021 (*Funded*)
- Title: "Closing the Si Cycle in Rice Agroecosystems to Sustainably Control As and Cd Uptake by Rice Grown Under Alternate Wetting and Drying (AWD)"
- Role: Post-award team member, data collection and analysis, field sampling campaign to Arkansas rice paddies, collaborative partnerships.

*Graduate Researcher / Project Contributor*, **Ministry of Environment of Korea** (USD 861K) 2019–2022 (*Funded*)
- Title: "Mitigation and Control of Low-Level $N_2O$ Emissions in Wastewater Treatment Plants (Global Top Environmental R&D)"
- Role: Contributed to the development of the full proposal, provision of seed data, lab-scale and pilot-scale biofilter operation, project management, training of members, collaborative partnerships.

## TEACHING AND MENTORING EXPERIENCE

***Teaching***:

Guest lecturer, **Seoul National University** May 2026
- Provided an invited lecture titled "Predicting PFAS Fate in Subsurface Environments via Biogeochemical Controls" and discussion section for course "*CEE 457.634: Contaminant Transport Analysis* (3 credits; 15 students)"

Teaching assistant, **Cornell University** Spring 2022, Spring 2023
- Teaching assistantship for course "*CEE 4530: Laboratory Research in Environmental Engineering* (3 credits; 15-20 students)"
- Led lab sessions, office hours, and student project management
- Delivered hands-on instruction on analytical instrumentation and data analysis for undergraduate researchers.

Instrument manager of Biogeochemistry and Ecological Eng. Lab, **Cornell University** 2021–2023
- Provision of regular training on the operation and maintenance of analytical instruments, including (HPLC-)ICP-MS, IC, and TOC analyzer

***Mentoring***:

Research mentor, **University of California, Berkeley** 2024–present
- Mentoring and training on essential protocols around PFAS analysis to Ph.D. students (Ananya Visweswaran, Nicole Garrido, Leyan Peng) and undergraduate researchers (Olivia Wang, Peter Reyes, Diego Canela, Albert Yu)

## PEER-REVIEW ACTIVITIES

Ad-hoc reviewer for:
- *Environmental Science & Technology* (16 reviews completed)
- Other peer-reviewed journals: *Applied and Environmental Microbiology, ACS Omega, Agrosystems, Geosciences & Environment*

Student presentation judge for:
- *Superfund Research Program (SRP) Annual Meeting 2026*

## INVITED TALKS

1. **Invited research seminar,** December 2026, *AEESP Future Faculty Seminar Series* | Host: Association of Environmental Engineering and Science Professors (AEESP) (scheduled)
2. **Invited lecture for course "*CEE 457.634: Contaminant Transport Analysis*",** May 2026, "Predicting PFAS Fate in Subsurface Environments via Hydrologic and Biogeochemical Controls," *Department of Civil and Environmental Engineering,* ***Seoul National University***, Seoul, South Korea | Host: Dr. Yongju Choi
3. **Invited research seminar,** April 2026, "Predicting PFAS Fate in Subsurface Environments via Hydrologic and Biogeochemical Controls," *Department of Civil and Environmental Engineering,* ***Korea Advanced Institute of Science and Technology (KAIST)***, Daejeon, South Korea | Host: Dr. Sukhwan Yoon
4. **Invited research seminar,** March 2026, "Redox and Soil–Water Chemistry Controls on Contaminant and Nutrient Dynamics," *Department of Biological and Agricultural Engineering,* ***Kansas State University***, Manhattan, KS | Host: Dr. Mark Wilkins
5. **Invited research seminar,** July 2025, "Regulation of Arsenic Methylation and Demethylation Dynamics by Organic Carbon in Rice Paddy Soil & Effects of Iron Minerals and Reductive Dissolution on PFAS Precursor Sorption in AFFF-Impacted Soils," *Department of Green Chemistry,* ***University of Chemistry and Technology, Prague (UCT Prague),*** Czech Republic | Host: Dr. Lenka McGachy

## CONFERENCE PRESENTATION

1. **Hyun Yoon**, John Han, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Machine Learning Reveals How Molecular Structure and Environmental Conditions Jointly Drive PFAS Biotransformation. ***Superfund Research Program (SRP) Annual Meeting 2026.*** Portland, OR, *poster*
2. **Hyun Yoon**, Anna-Ricarda Schittich, Fuhar Dixit, Jonathan J. Uhler, Leyan Peng, Fanshu Geng, Damian Helbling, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Charge-Dependent Sorption Controls the Bioavailability and Biotransformation of PFAS Precursors in AFFF-Impacted Solids. ***Gordon Research Conference 2026 Environmental Sciences: Water.*** Plymouth, NH, *poster*
3. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2025. How organic matter and iron mineral dynamics, including reductive dissolution, affect PFAS sorption to AFFF-contaminated solids. ***Berkeley Energy & Resources Collaborative (BERC) 2025 Resources Symposium: Water as a Resource***. Berkeley, CA, *poster*
4. Michael Vega, **Hyun Yoon** and Matthew C. Reid. 2025. Regulation of arsenic methylation and demethylation dynamics by organic carbon in rice paddy soil. ***Goldschmidt 2025 Conference***. Prague, Czech Republic, *oral*
5. **Hyun Yoon**, Anna-Ricarda Schittich, Leyan Peng, Nicole Garrido, Fuhar Dixit, Christopher Olivares, David Sedlak and Lisa Alvarez-Cohen, 2024. Assessment of physical, chemical, and biological factors controlling biotransformation of cationic and zwitterionic precursors in PFAS source zones. ***DOD Energy Innovation Symposium***. Washington, DC, *poster*
6. **Hyun Yoon**, Fuhar Dixit, Katerina Tsou, Yanghua Duan, David L. Sedlak and Lisa Alvarez-Cohen. 2024. Effects of iron, organic matter, and seawater intrusion on PFAS precursor sorption in soil and groundwater. ***SETAC North America 45$^{th}$ Annual Meeting***. Fort Worth, TX, *oral*

7. **Hyun Yoon**, Benjamin Stenzler, Lena Abu-Ali, Maria P. Asta, Alexandre J. Poulain and Matthew C. Reid. 2023. Using an anaerobic biosensor to model the effects of dissolved organic matter and complexed ferric iron on arsenite bioavailability. ***ACS Fall 2023 National Meeting***. San Francisco, CA. *oral*
8. **Hyun Yoon** and Matthew C. Reid. 2023. What is the role of carbon catabolite repression in regulating microbial uptake and methylation of arsenic? ***2023 AEESP Research & Education Conference***, Boston, MA. *poster*
9. **Hyun Yoon**, Martin P. Pothier, Xuhui Zhang, Andrea Giometto, Alexandre J. Poulain and Matthew C. Reid. 2021. Kinetic study of microbial arsenite uptake using whole-cell biosensor: Inhibitory effects of thiols and low molecular weight organic molecules. ***ACS Spring 2021 National Meeting***, Virtual meeting. *oral*
10. **Hyun Yoon**, Min Joon Song, Daehyun D. Kim and Sukhwan Yoon. 2018. Serial biofiltration system for effective removal of low-concentration nitrous oxide in oxic gas streams. ***17$^{th}$ ISME***, Leipzig, Germany. *poster*
11. **Hyun Yoon** and Sukhwan Yoon. 2017. Removal of low concentration nitrous oxide from gas streams using a self-sustaining biofiltration system with gravitationally-fed wastewater as the nutrient solution. ***ASM Microbe 2017***, New Orleans, LA. *poster*

## PATENT

1. Sukhwan Yoon, **Hyun Yoon**, Min Joon Song, Heejoo Han and Daehyun D. Kim. Apparatus for removing nitrous oxide by using biofilters. ***Korea Patent*** No. 10-2227395-0000, filed Apr 2019 and issued Mar 2021.
2. Sukhwan Yoon and **Hyun Yoon**. The self-sustaining biofiltration apparatus for removal of nitrous oxide and the method of its installation and operation at wastewater. ***Korea Patent*** No. 10-1923992-0000, filed Jan 2017 and issued Nov 2018.

## LEADERSHIP & ACTIVITIES

Graduate Student Association of Civil and Environmental Engineering, **Cornell University** 2020–2021
- Role: Social media chair

Glee Club (Choir), **Yonsei University** 2013
- Role: Baritone, Club executive

Squad Leader as Tank Driver, **Republic of Korea Army** 2011–2012
- Rank at Discharge: Sergeant
- Led a squad comprising three tanks and 12 soldiers